

Dynamic Pricing for European Camping Group

Jimmy Zhang, Tobias Vanoverschelde, Fausto De Valck, Andrew Posey, Alexander Lievens
KU Leuven, Statistical Consulting 2025-2026, ORTEC Project

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1 Problem and Business Context

European Camping Group (ECG) operates over 400 campsites across 10 European countries with several accommodation types and four distinct market segments. With a 30-week open season, this is more than 250,000 individual prices to set every year, one for every combination of campsite, accommodation, market, and arrival-week. We refer to such a combination as a *ROMGID* (reservable-option-market-group), the unit of analysis throughout the report. Based on the information and business case provided, this report will examine the following question. Can historical booking data be used to recommended better prices for ROMGID's and increase ECG's revenue?

2 The Trap

The most direct approach, regressing booked nights on price while controlling for accommodation and seasonality, yields an answer that is economically impossible. On a 50,000 row training sample, the estimated price coefficient is positive: $\beta = +0.29$, with an 80% credible interval of $[+0.18, +0.42]$. Read literally, this means “raising prices increases bookings”.

The reason is structural, not statistical. ECG's pricing system already reacts to demand: when a campsite fills up, prices rise, when bookings slow down, prices fall. Due to this, high prices and high bookings tend to occur together in the data, not because price causes bookings, but because both respond to a common, unobservable underlying demand signal.

The fix requires identifying this demand signal that the pricer reacts to. Bookings on books (BoB), the total number of nights a ROMGID has previously been reserved when a price is set. Including this variable in the model helps remove historical booking bias and better isolate the true impact of price on individuals' bookings.

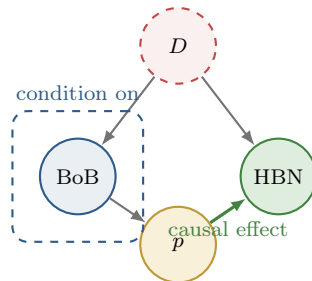


Figure 1: Causal structure linking price (p) and bookings (HBN). Latent demand D (red, dashed, unobservable) drives bookings-on-books (BoB , observed) and bookings directly. The existing pricer sets p in response to BoB . The path $p \leftarrow BoB \leftarrow D \rightarrow HBN$ is a problem: it lets demand contaminate the apparent price-bookings relationship. Conditioning on BoB (dashed blue box) removes this, leaving only the genuine causal effect $p \rightarrow HBN$.

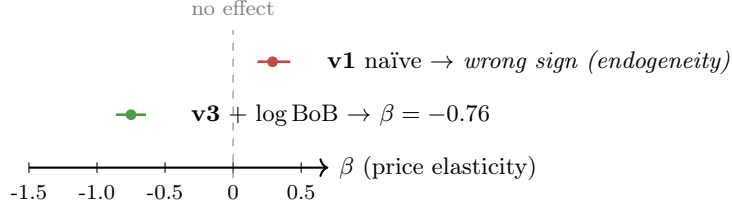


Figure 2: Posterior point estimate and 80% credible interval for the global price elasticity β . The naïve specification (v1) proposes an economically impossible impact, direct evidence of the correlation between price and bookings in ECG’s data. After conditioning on bookings-on-books, the elasticity becomes negative as economic theory requires (v3, $\beta = -0.76$).

3 Two Models, Two Jobs

The Bayesian estimate of $\beta = -0.76$ from model v1 implies demand is inelastic on average ($|\beta| < 1$). This means that for every 1% increase in price, bookings decrease by around 0.76%. Because this response is less than proportional, total revenue (price multiplied by bookings), on average, goes up when prices go up, within the price range our data cover. Under a constant-elasticity assumption, this would mean ECG could raise prices and capture more revenue. Based on this learning, we created an improved gradient-boosted-trees model (LightGBM) optimiser, referenced in Section 4, that helps clarify our understanding. Average elasticity across ROMGIDs is correct, but the per-ROMGID optimal price is varied. Some sites are overpriced, some are underpriced, and the total sum of movements is close to zero. Therefore, the pricing engine should focus on specific sites, rather than all sites collectively, to adjust prices on in order to maximize revenue.

Model v3 is intentionally minimal as it only uses the variables strictly needed to remove the impact of BoB. Through this minimalism, it is an effective estimator of the local average elasticity, but it is unable to make future predictions. To predict bookings at a specific time and price, we trained a LightGBM model on the same conditions as before but included lead-time and calendar and accommodation features that were intentionally left out of v3. Trees were chosen for this model because they will not overfit when multiple variables are included unlike some Bayesian models. At typical prices, this LightGBM model produced the same beta as the Bayesian model (around -0.75). This brings validity to our elasticity value as two different models with different assumptions resulted in the same answer.

Out-of-sample, LightGBM achieves a median ROMGID-level error of 27% on total booked nights, which is within reason for the industry. This error is concentrated in low-volume ROMGID’s where small absolute differences in bookings inflate percentage errors. For ROMGID’s with at least 10 bookings per season, the median error drops to 22%.

4 The Optimiser and What It Recommends

For each tested snapshot independently, we ask the LightGBM forecaster a question, at this point in the booking horizon, what price would have produced the highest predicted revenue? We simulated 21 different price points, ranging between up to 15% higher or lower than historical rates, to determine the exact price that maximises predicted revenue. Therefore, for every ROMGID, the output is 53 prices across the total booking horizon, one per snapshot.

Three constraints keep the recommendation logical. The potential range is capped at $\pm 15\%$ of historical prices for each ROMGID, predicted bookings are can only reach capacity and nothing further, and every recommendation passes through a two-tier flag (green / yellow) system before deployment, keeping humans in the loop.

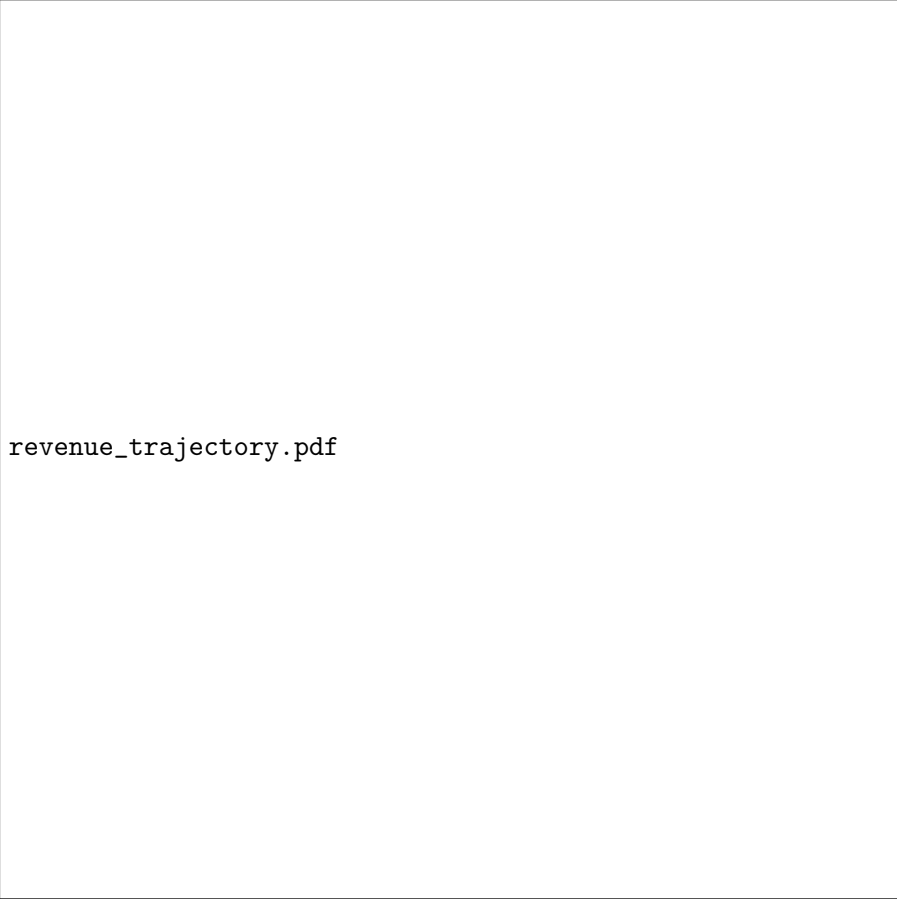
Applied to 14,200 ROMGID's in the held-out test fold:

Metric	Value
Median per-ROMGID Δ price	-0.7%
Δ price IQR (per ROMGID)	$[-6.0\%, +5.4\%]$
% snapshots at upper bound ($1.15\times$)	12.7%
% snapshots at lower bound ($0.85\times$)	14.7%
% snapshots with interior optimum	72.6%
Total predicted uplift	€63M
Uplift as % of baseline	+38.8%
Flag split (green / yellow)	51% / 49%

recommendations_distribution.pdf

Figure 3: Distribution of recommended price changes across all test-fold snapshots. The histogram is roughly symmetric around zero, with mass at both the lower and upper safety bounds. About 73% of decisions land at an interior optimum within the safety bounds, 14.7% land at a 15% discount, and 12.7% land at a 15% increase.

The most important number on the above figure is the **72.6% interior-optimum rate**. For nearly three quarters of snapshot decisions, the model finds a revenue-maximising price strictly inside the $\pm 15\%$ safety bounds, not at a boundary. This is something that the constant-elasticity Bayesian model in Section 2 could not produce. When elasticity is forced to be constant, the revenue function has no interior maximum since an increase in price will always increase revenue, but when LightGBM develops a flexible price-demand curve, genuine interior optima emerge. The recommendations are also bidirectional as both price increases and decreases are promoted. From this model, ECG is overpricing some cells and underpricing others, with the majority of decisions sitting at modest price adjustments.



revenue_trajectory.pdf

Figure 4: Predicted revenue per snapshot for one representative ROMGID, observed price (blue) versus recommended price (amber), with 80% prediction intervals from a quantile LightGBM. Most of the booking horizon shows similar predicted revenue under both strategies, with the recommended trajectory slightly higher in the mid-horizon weeks where the majority bookings actually concentrate.

The +€63M total predicted uplift is a model projection and is likely optimistic for three reasons. First, the model has a 27% out-of-sample error on observed prices, and utilizing predicted prices pushes it further from historical data. Second, the optimiser picks the candidate that maximises predicted revenue, exploiting any potential model error that may occur. Third, the model can predict on small positive bookings in late-horizon snapshots, where historical bookings are almost zero, and then sees these bookings as small revenue gains. While the directional adjustments of the model are robust, the magnitude of the price changes is not. Therefore, a randomised price experiment, discussed in the next section, is the best way to improve the validity of the model for ECG.

5 What ECG Should Do Next

Two complementary tracks follow from this report. The first validates the magnitude. The second deploys the workflow.

1. Run a small randomised price experiment. The +€63M figure is a model projection, not a calibrated lift estimate. The three reasons it is optimistic, listed at the end of Section 4, cannot be resolved by further analysis of the historical data. They can only be resolved by generating new data under a known, randomised intervention. We recommend selecting 100 to 200 ROMGID's across markets and accommodation types, randomising their price by $\pm 25\%$ over a single 30-week season, and observing the resulting bookings. This delivers two things at once: an unbiased elasticity estimate to refit the model on, and an extension of the recommendation envelope beyond the $\pm 15\%$ safe-range cap

that currently bounds the optimiser. The cost is bounded uncertainty on a small subset of inventory; the return is a deployable, calibrated pricing system.

2. Run the pipeline in shadow mode alongside the experiment. The pipeline produces a `recommendations.csv` every time it runs, with one row per ROMGID containing the current price, the recommended price, predicted bookings, revenue uplift with an 80% credible interval, and a human-in-the-loop flag. Because the model conditions on bookings-on-books at the moment of prediction, any new reservation that enters ECG’s booking system automatically shifts the ROMGID to a different point on the demand curve. A nightly batch job recomputes features for all active ROMGID’s, reruns the price sweep, and regenerates the file, so the dashboard always reflects the most recent booking state without requiring model retraining. Retraining itself happens once per season, when a full new round of observations is available to update the elasticity estimates.

The prototype dashboard we deliver alongside this report reads the recommendation file and presents each ROMGID as a row in a sortable, filterable table. A pricing manager opening the dashboard sees, for every campsite-week combination, the current price next to the model’s recommended price, the percentage change, the predicted revenue uplift, and a colour-coded flag. Clicking a row expands a detail panel with observed versus predicted bookings, a capacity-utilisation bar, historical price volatility, and a plain-language explanation of why this recommendation was flagged the way it was.

ROMGID	Campsite	Type	Flag	Current	Recommended	Δ price
0042	CS-FR-118	Mobile Home	yellow	€187	€164	−12.3%

Figure 5: Example row from the pricing dashboard. This ROMGID is flagged yellow because the recommended adjustment exceeds 10%; the pricing manager reviews local context before approving.

The flag system connects the model to the human. A **green** flag means the recommended adjustment is small (at most 10%) and the historical price has been stable (coefficient of variation below 0.15). A **yellow** flag means the adjustment is larger (10–20%) or the historical price has been moderately volatile; the model’s recommendation is plausible but needs a manager to check what the model cannot see, a local festival, an incoming weather front, a competitor’s flash sale. Yellow recommendations form the manager’s daily review queue.

Why Both Tracks are Necessary. The two tracks answer different questions. The experiment validates the *magnitude* of the recommended adjustments by generating randomised counterfactual data the historical record cannot provide. Shadow mode builds the *managerial trust* required for adoption, by letting the team compare the engine’s suggestions against their own decisions over a season and develop a feel for where the model adds value and where domain expertise remains essential. Selective auto-application of green-flag recommendations should follow only once both signals are positive: the experiment confirms the elasticity at the ROMGID level, and shadow-mode track records show the model is making sensible recommendations in segments managers trust. Until the experimental data are in, every recommendation remains advisory. The manager’s role then shifts from setting every price to auditing the exceptions, a more efficient use of their domain knowledge.

Limitations. The methodology rests on three assumptions worth flagging. First, we close one back-door (demand-autocorrelation via log BoB). Other sources of price endogeneity, such as channel-mix changes and manager overrides, remain unobservable in this dataset. Second, LightGBM’s 27% out-of-sample error grows for counterfactual prices the model has not seen, and the optimiser exploits that error in its own favour. Third, capacity is structurally non-binding here (median utilisation 10%); a real deployment in peak weeks at premium properties would encounter the capacity constraint that is currently a tested but inactive safeguard.

6 Conclusion

Treating dynamic pricing as a causal-inference problem reveals a clear endogeneity trap and a defensible local elasticity of -0.76 once it is closed. A complementary LightGBM forecaster delivers per-snapshot recommendations that are differentiated and bidirectional, identifying which cells should move and in which direction. The outcome is not just a model but a working pipeline: a recommendation file, a flag-based review workflow embedded in a dashboard, and a phased integration plan in which a one-season randomised experiment validates the lift magnitude while shadow-mode operation builds managerial trust. The engine does not replace the pricing manager; it focuses their attention on the ROMGID's where intervention matters most, while preserving their existing expertise rather than overruling their judgement entirely. This aspect is also important in convincing the existing managers to adopt the project rather than perceiving it as a threat to their role.

ECCG's status quo is likely leaving model-identifiable revenue on the table. The size of that revenue is the open question this report cannot answer alone, but a small randomised experiment, run alongside shadow-mode deployment of the pipeline, is designed to answer it within one season at bounded cost.